

Figure 1: Last-iterate $\mathrm{SubOpt}(\hat{\mu})$ as a function of the number of observed poacher actions. In (a), a random leader policy is either fixed for the trajectory or, for Online random, redrawn uniformly from the simplex after every observation. Warm and no-warm Active use information multipliers $\lambda/\eta^2 = 0.003$ and 0.03, selected on transfer tasks and frozen before final evaluation. In (b), MLE observes follower actions under the source Stackelberg-optimal leader, whereas the OLS methods use a joint-QRE expert required by their equilibrium equations. Zero-sum OLS additionally observes one expert leader action per poacher action, so its total observed-action count is twice the horizontal-axis value. Shading denotes pointwise 95% confidence intervals over 50 seeds, clipped at the axis limits where they extend below the plotted range.

1 Numerical validation

We compare reward-learning protocols in a GUARD-derived Lobéké game with a simulated logit-response follower. Active methods choose the patrol policies used to collect observations; fixed-expert methods receive demonstrations from a source-task equilibrium oracle.

Every evaluation leader reward is represented positive-affinely in $[0, 1]$, so that $R_{\max} = 1$. This leaves all leader-policy orderings unchanged, and the reported metric remains ordinary SubOpt.

The retained experiments measure source-task performance and transfer to new leader objectives without further follower observations. Warm-up and no-warm-up Active IRL use different information multipliers, so their comparison alone does not isolate the effect of warm-up.

These results concern one fixed game and a correctly specified response model. The selected downstream objectives illustrate transfer within that setting; they are not a sample of independent environments.

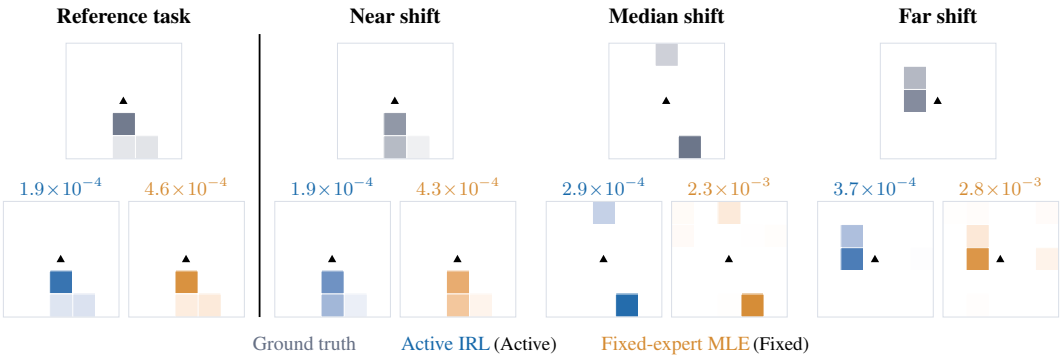


Figure 2: **Patrol coverage and associated mean** $\mathrm{SubOpt}(\hat{\mu})$ **across downstream tasks.** Transfer after 1,600 observed poacher actions. The follower reward is unknown to Active IRL, which is compared with an MLE fitted to observations under the source Stackelberg-optimal leader. Both estimators use the same logit follower model, but only Active IRL selects leader policies to learn a reward that transfers across objectives. The three illustration tasks are selected from the predeclared ten-task suite as the nearest, median, and farthest behavioral shifts, measured by total variation between the source and downstream true optimal leader policies (0.23, 0.86, and 1.00). This selection does not inspect any learned estimate or method performance. Values are task-specific mean $\mathrm{SubOpt}(\hat{\mu})$. Darker cells indicate greater coverage; the triangle marks the ranger base.



Figure 3: Alternate layout of the downstream patrol comparison.

A ADDITIONAL EXPERIMENTS

A.1 LOBÉKÉ ENVIRONMENT FROM THE GUARD BENCHMARK

Task and observations. We use the Lobéké instance generated from GUARD's anti-poaching data: a 5×5 grid with 23 accessible targets and 23 ten-step patrol actions. The ranger commits to a mixed patrol policy, and a simulated logit-response poacher chooses a target. One interaction supplies one observed poacher action. Zero-sum OLS additionally observes one sampled ranger action per interaction. The benchmark therefore compares protocols with different observation assumptions.

The source reward is zero-sum before reporting normalization, $r^1_{\mathrm{src}} = -r^2_{\omega^\star}$. We report each leader reward in $[0, 1]$:

$$\bar{r}^1(a, b) = \frac{r^1(a, b) - \min_{a',b'} r^1(a', b')}{\max_{a',b'} r^1(a', b') - \min_{a',b'} r^1(a', b')}.$$

This preserves optimal policies and makes SubOpt comparable on a unit reward scale. During estimation, the source payoff is used by the evaluator and demonstration oracle, not supplied as an input to the reward estimator. The estimators fit observed actions; they do not recover the follower reward by negating that payoff. If the unnormalized source reward and its exact zero-sum relationship were both supplied to a learner, the follower reward would already be known without observations.

Transfer tasks. Transfer changes the ranger objective while holding the park, follower reward, and response model fixed. Let $c_{ab} \in [0, 1]$ denote coverage of target b by patrol a. For signed priorities $z \in [-1, 1]^{23}$, the unnormalized reward is

$$r^1_z(a, b) = (2c_{ab} - 1)z_b.$$

The ten-task evaluation suite uses seed 9200. We generate 500 independent uniform priority vectors and greedily select ten diverse payoff matrices: start with the first candidate, then minimize the largest absolute cosine similarity with previously selected matrices. Normalize selected rewards to $[0, 1]$ and weight them equally. Reported confidence intervals are over learning seeds, conditional on this fixed synthetic task suite.

The retained hyperparameters came from an earlier ridge-10^{-4} selection stage using eight learning seeds and 100 tasks generated with seed 9400. The later ridge-zero grid reuses that selection split for sensitivity analysis. The 50 evaluation seeds had also appeared in earlier development runs; correcting their estimator and reward normalization does not make them a fresh holdout. The separate prospective confirmation is documented with its own frozen protocol and does not replace these retained results.

The three downstream maps use the nearest, median, and farthest true optimal leader policies within the ten-task suite, measured by

$$D_{\mathrm{TV}}(z, \mathrm{src}) = \tfrac{1}{2}\|\mu^\star_z - \mu^\star_{\mathrm{src}}\|_1.$$

Task selection does not inspect learned estimates or method scores. The appendix heatmap uses all 180 priorities drawn with seed 9600, without diversity selection, grouped into terciles of the source-policy value gap

$$D_{\mathrm{gap}}(z, \mathrm{src}) = V_z(\mu^\star_z) - V_z(\mu^\star_{\mathrm{src}}).$$

These groups describe displacement from the source objective, not universal levels of learning difficulty.

Follower model. Let v_b be GUARD's value for an uncovered target. The raw features and utility are

$$\phi_{\mathrm{raw}}(a, b) = (c_{ab}, v_b, c_{ab}v_b), \qquad r^2_\omega(a, b) = \langle \phi_{\mathrm{raw}}(a, b), \omega \rangle.$$

The generated utility matrix yields $\omega^\star = (0, 1, -2/3)$ to numerical precision. The simulated follower has known inverse temperature $\eta_{\text{raw}} = 1.25$. In code, features are divided by $L = \max_{a,b} \|\phi_{\text{raw}}(a, b)\|_2$ and inverse temperature is multiplied by L, giving $\eta \approx 2.165$. Thus $\phi = \phi_{\text{raw}}/L$ and $\eta = L\eta_{\text{raw}}$ preserve every response probability:

$$\nu_{\mu,\omega}(b) = \frac{\exp(\eta \sum_a \mu(a)\langle\phi(a, b), \omega\rangle)}{\sum_{b'} \exp(\eta \sum_a \mu(a)\langle\phi(a, b'), \omega\rangle)}.$$

Only ω is learned. These experiments test a correctly specified simulated response model; they do not establish robustness to unknown temperature or real poacher behavior.

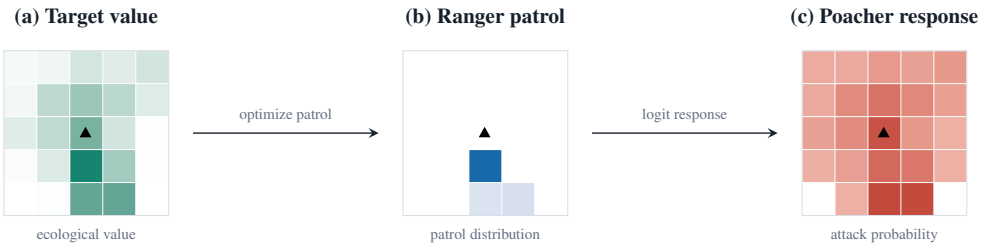


Figure 4: Rewards, patrols, and poacher behavior in the Lobéké environment. The 23 accessible poaching targets are embedded in a 5×5 park grid; darker cells indicate larger values within each map, and the triangle marks the ranger base. The source task is zero-sum: the ranger reward is the negative of the poacher utility. A mixed ranger policy induces target-level coverage probabilities, and the bounded-rational poacher responds to their expected utilities according to the logit model.

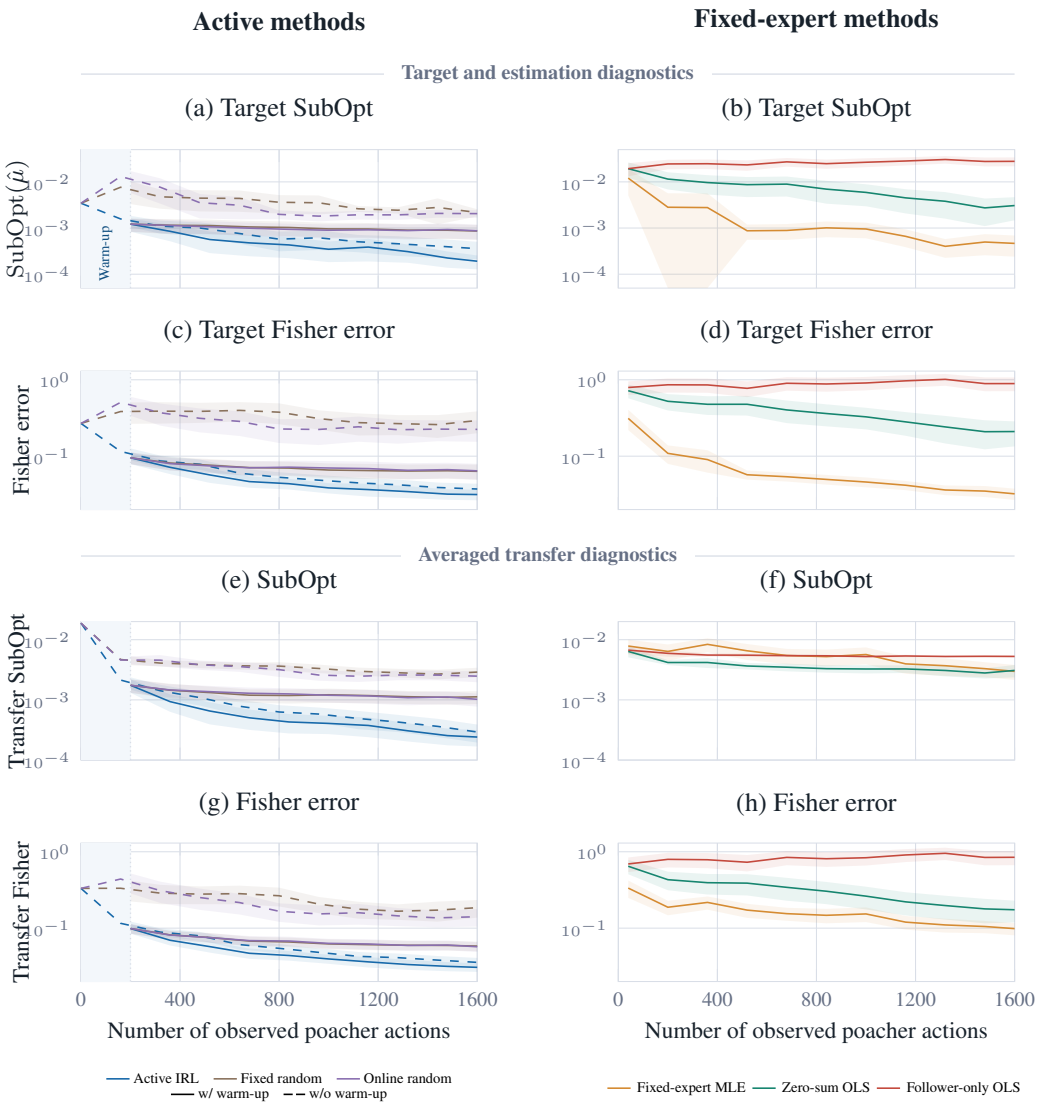


Figure 5: Target, estimation, and averaged-transfer diagnostics as a function of the number of observed poacher actions. Columns separate the active-design and fixed-expert data-access assumptions.

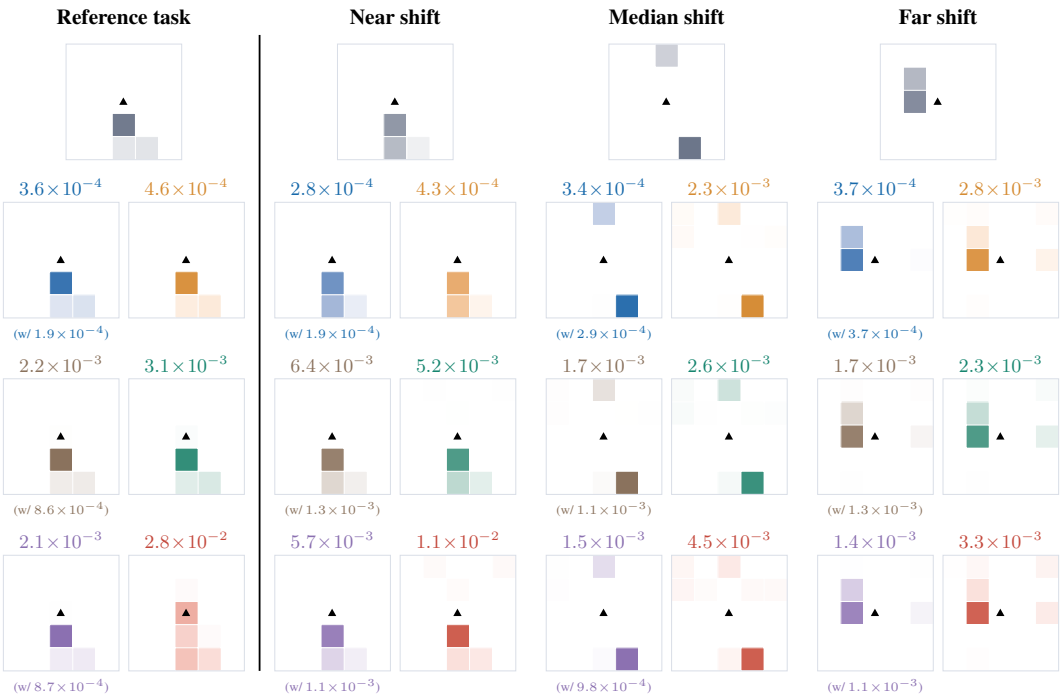


Figure 6: Patrol coverage under both data-access assumptions. Each task column pairs an active-design method on the left with a fixed-expert estimator on the right; the two halves are independent methods, not a matched comparison. Active IRL, Fixed random, and Online random estimate the poacher reward from leader policies selected by their own exploration protocols and assume no access to a source-task expert. Fixed-expert MLE observes poacher actions under the source Stackelberg-optimal leader and retains the general logit likelihood; both OLS estimators instead require a fixed entropy-regularized joint-QRE expert, with zero-sum OLS additionally assuming a zero-sum source game and observing one expert leader action per poacher action, and follower-only OLS using only the follower equations. Every method is shown in its no-warm-up configuration, the only one the fixed-expert estimators admit; the parenthesised value below each active-design map is the same method's SubOpt with warm-up, whose recovered policies are visually indistinguishable. The downstream columns are the nearest, median, and farthest behavioral shifts from the predeclared ten-task suite, selected solely by total variation between true optimal leader policies and independently of algorithm performance.

Figure 5 uses positive-affine reward representations in $[0, 1]$, so that $R_{\max} = 1$, without changing leader-policy orderings. The left column compares active-design protocols that may choose the leader policy used to elicit each poacher observation; the right column compares fixed-expert estimators that instead assume observations from a source-task expert. Corresponding rows use identical axes, while the separate legends keep these data-access assumptions explicit. Below the labeled divider, transfer metrics are averaged taskwise over a fixed deterministic suite of ten downstream leader rewards and then across 50 evaluation seeds; bands are 95% confidence intervals conditional on that task suite. The pale region marks the first 200 observations used by warm-up variants.

A.2 Performance in synthetic environments

A.2.1 Sparse-information game

A.3 Active IRL on the lower-bound instance

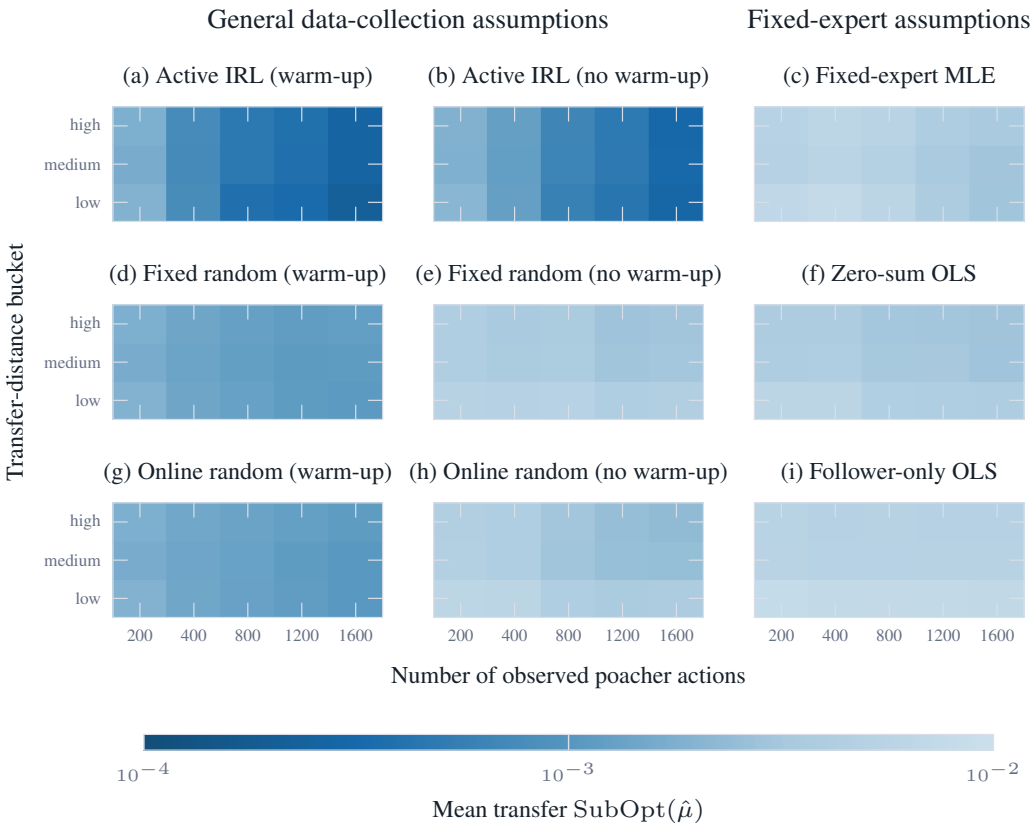


Figure 7: Transfer SubOpt by the downstream value gap of the true source-optimal leader policy. The 180 normalized downstream rewards and their tie-aware buckets are fixed without reading any method output; each cell averages every task in its bucket and all 50 learning seeds. Gap is smooth enough to form exactly balanced $60/60/60$ terciles. Columns separate methods with general data-collection access from methods requiring a fixed source expert, and color is logarithmic.

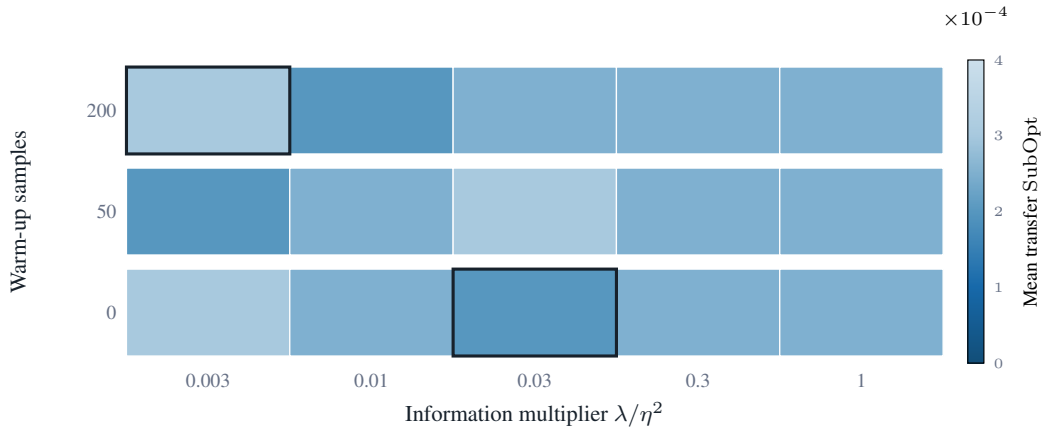


Figure 8: Estimator-matched post-freeze sensitivity of Active IRL to the warm-up budget and information multiplier after 1,600 observed poacher actions. Colors show mean downstream $\mathrm{SubOpt}(\hat{\mu})$ over 100 fixed normalized tasks and eight paired audit seeds. Equal-width columns represent the five discrete settings shown on the horizontal axis; darker cells are better and the color scale is fixed to $[0, 4 \times 10^{-4}]$. Zero warm-up samples denotes the no-warm-up design, and outlined cells are the final protocols frozen before this ridge-zero audit. For every other cell, the paired 95% confidence interval for the difference from the corresponding frozen protocol contains zero; this means unresolved, not equivalent. These descriptive intervals are not multiplicity-adjusted. The audit seeds and tasks are disjoint from final evaluation, and the audit did not reselect the final protocols. A 20-seed, 100-task ridge-zero replay of an earlier confirmation split gives lower mean transfer SubOpt for both retained protocols than fixed-expert MLE. It reuses previously examined seeds and tasks; it is not a new independent confirmation. Matched warm-up contrasts remain unresolved.

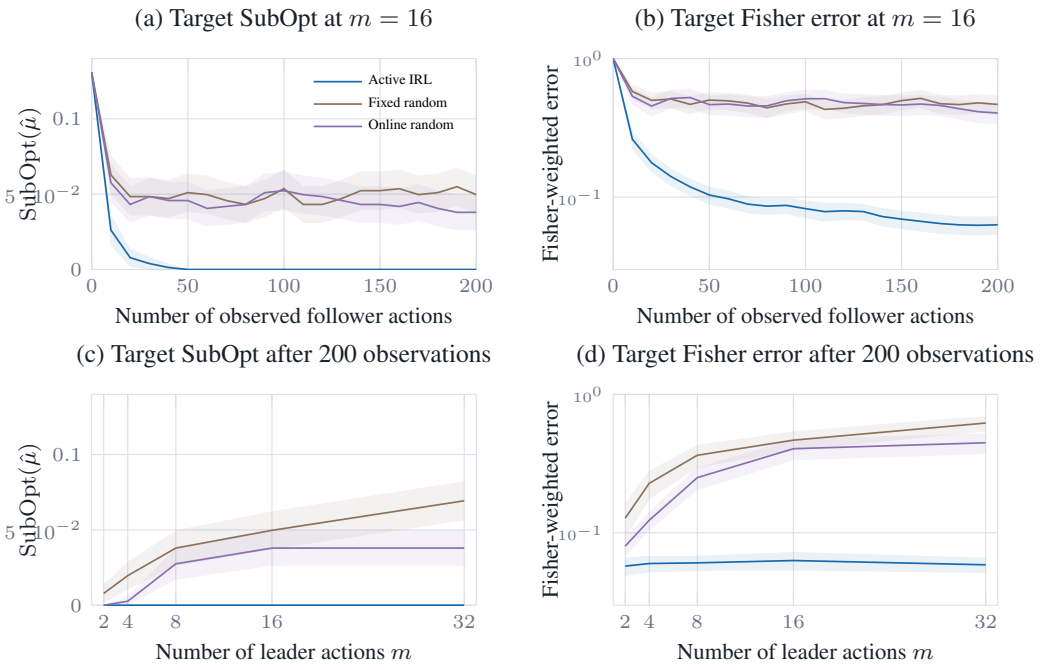


Figure 9: Target SubOpt and Target Fisher error in a game with one informative pure leader action. Mixed policies remain informative whenever they assign positive mass to that action; uniform-simplex random policies typically assign it less mass as the number of actions increases.

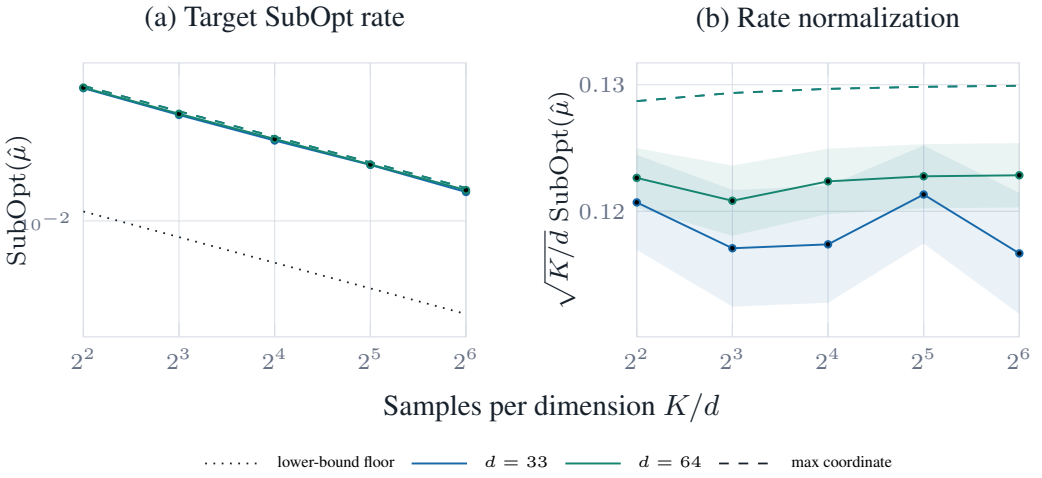


Figure 10: Empirical behavior on the minimax hard family. The reward separation decreases with the observation budget, $\epsilon = \frac{1}{2}\sqrt{d/K}$, so even a policy that ignores observations has SubOpt of order $\sqrt{d/K}$. This is an illustration of the family, not empirical proof of a minimax guarantee or a convergence rate on one fixed instance. Each budget and dimension includes 16 independent replicates of every active coordinate.